

University of Trento
Master's Degree in Computer Science



**UNIVERSITÀ
DI TRENTO**

Data Visualization Technical Report

Data Visualization Lab
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Davide Donà	264467
Andrea Blushi	264468
Lorenzo Di Berardino	265984

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1 City Bike Dataset

For this project, our main dataset is the Citi Bike system dataset, a public repository tracking New York City bike trips since 2013.

1.1 Dataset Structure

Released monthly as open-data CSV files, these records log every completed ride, with high-volume months split across multiple files. Each row represents a single trip, offering detailed attributes that characterize the journey.

1.1.1 Original Structure

The original dataset includes the following columns:

- **Ride ID:** A unique identifier for each bike trip.
- **Rideable Type:** The type of bike used for the trip (“classic bike” or “electric bike”).
- **Started At:** A timestamp indicating when the trip started.
- **Ended At:** A timestamp indicating when the trip ended.
- **Start Info:** Information about the station and location where the trip started, structured as:
 - **Station Name:** The name of the starting station.
 - **Station ID:** The identifier of the starting station.
 - **Latitude:** The latitude coordinate of the starting station.
 - **Longitude:** The longitude coordinate of the starting station.
- **End Info:** Information about the destination station, structured equally to Start Info.
- **User Type:** The type of user (“member” or “casual”).

1.1.2 Extracted Features

Additional per-trip features were derived to support the analysis:

- **Trip Duration:** The duration of the trip in seconds, calculated as Ended At – Started At.
- **Trip Distance:** The distance between the start and end stations, computed using the Haversine formula and multiplied by a circuitry factor of 1.3 to better approximate real routes.
- **Day of the Week:** The weekday extracted from the end timestamp, which is used to partition the data since trips are published based on when they ended.
- **Hour of the Day:** The hour extracted from the end timestamp, for the same reason.

Beyond these per-trip features, two aggregated datasets are derived from the cleaned trip data to support station- and route-level analysis:

- **Station Popularity:** for each station and time period, the number of trips starting and ending at that station, obtained by grouping trips by station.
- **Trips Flow:** for each pair of stations and time period, the number of trips travelling between them, obtained by grouping trips by station pair.

1.2 Analysis and Insights

To ensure transparency and reproducibility, the entire analysis is supported by Jupyter notebook within our repository. The exploratory analysis is based on the February 2025 trip file and focuses on temporal, spatial, and behavioural patterns in bike usage.

Temporal Analysis Temporal analysis shows clear variations in demand depending on the hour of the day and the day of the week. Trips are concentrated during commuting hours, while weekends show different usage patterns associated with leisure activities.

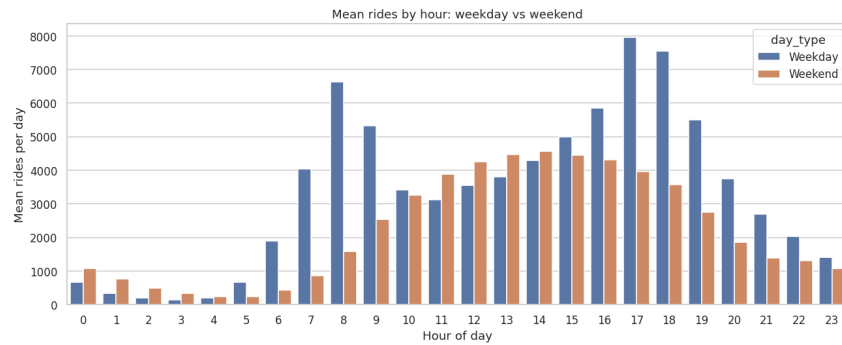


Figure 1: Distribution of trips by hour of the day and weekday/weekend.

Spatial Analysis Spatial analysis highlights the distribution of station origins across the city. Station popularity and trip flows reveal that demand is concentrated around Manhattan and other high-density areas, with some stations acting as important origins.

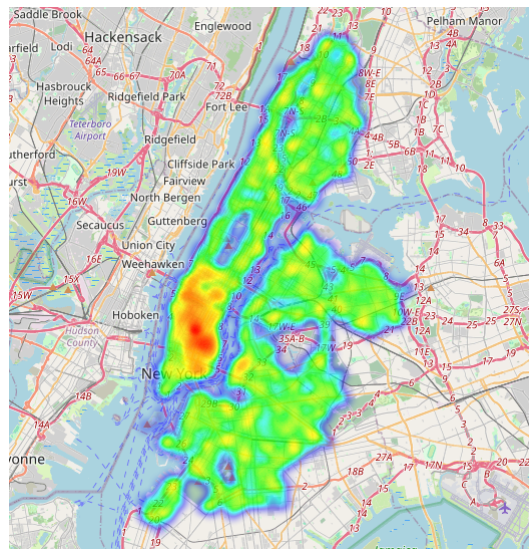


Figure 2: Distribution of trips by station of origin, showed through an heatmap.

User/Bike Type Behaviour Analysis As shown in Figure 3, casual riders lean much more heavily on electric bikes than members do. Furthermore, casual riders take longer trips on both bike types, whereas members maintain a steady, short duration regardless of the bike they choose.

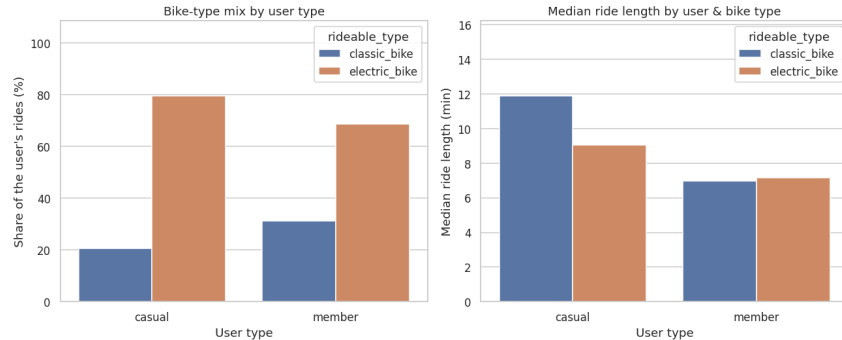


Figure 3: Distribution of trips by user/ride type based on shares of rides and trip length.

Trip Duration and Distance Trip duration and distance distributions provide additional insight into mobility behaviour. Most trips are relatively short, while longer rides are less frequent and often correspond to recreational usage. Durations begin at one minute, as Citi Bike removes trips shorter than 60 seconds at the source. The distributions of trip duration and distance shows some anomalies that we are describing in the following section.

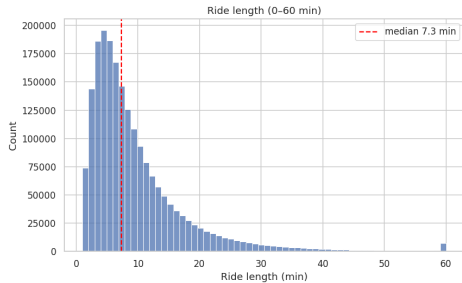


Figure 4: Distribution of trip durations.

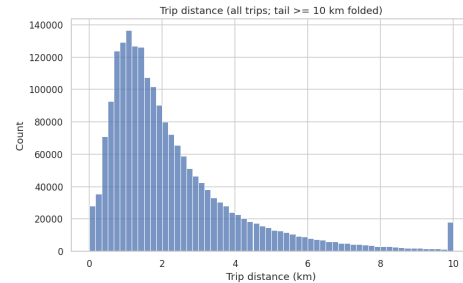


Figure 5: Distribution of trip straight-line distances.

These analyses justify our dashboard proposal, as they clearly reveal patterns in temporal demand, station-level activity, user and bike-type behaviour, and trip duration and distance.

1.3 Data Quality

Various data quality issues were encountered during the analysis of the City Bike dataset.

Different Legacy Structure Around 2020, Citi Bike updated its trip-file schema. Files from the 2013-2019 period use a legacy layout featuring different column names, dropped attributes, and distinct formatting (see the analysis notebook for further details). Despite these discrepancies, this historic data remains highly valuable.

To leverage it, the preprocessing pipeline implements specific schema adjustments, standardizing all monthly files into a single, compatible feature set.

Missing Station Values A small fraction of trips contain missing values, exclusively within the station-related columns. The end-of-trip fields are the most affected: in a representative month, roughly 4,800 trips (0.24%) lack an end station identifier, while start station fields are missing just 0.03% of the time. This issue likely occurs when bikes are abandoned or returned outside a dock. A complete per-column breakdown is available in the analysis notebook. Because valid origin and destination data is essential for spatial and distance metrics, these incomplete rows are dropped during preprocessing.

Trip Outliers Beyond missing values, a small number of trips are clear outliers. In a representative month, about 315 trips (0.02%) last up to 25 hours, appearing in the final bin of the duration histogram (Figure 4). These likely reflect unreturned bikes rather than continuous rides, so trips exceeding a maximum-duration threshold are dropped. Additionally, around 27,000 trips (1.4%) start and end at the same station, resulting in a zero straight-line distance (Figure 5); because distance metrics are meaningless for these round trips, they are removed. Finally, some trips report impossible distances over 11,000 km, as an example, due to corrupted station coordinates, and are filtered out using a maximum-distance threshold.

Biases This dataset carries certain inherent limitations. Because it relies exclusively on the Citi Bike system dataset, the findings may not accurately reflect the demographics of the entire population of cyclists in New York City. Additionally, trip distances are calculated using the Haversine formula between the start and end stations; this straight-line metric serves only as a geometric approximation and cannot account for the actual routes or detours taken by riders. Unfortunately, these structural data constraints cannot be resolved within the scope of our project.

2 Station Metadata Dataset (GBFS)

To enhance the station data provided by Citi Bike, we incorporate the real-time feed published in General Bikeshare Feed Specification (GBFS) format and documented on the Citi Bike System Data page.

2.1 Dataset Structure

Unlike historical trip records, station metadata is served via a public JSON API adhering to MobilityData’s GBFS standard. From this service, we query two key Citi Bike endpoints:

- `station_information.json`: Static properties like location and capacity, whose content changes only when the network itself changes.
- `station_status.json`: Live availability and operational state.

2.1.1 Original Structure

The two endpoints share the `station_id` field, on which they are merged to produce a unified station record comprising:

- **From `station_information`:**

- **Station ID:** internal GBFS identifier, a UUID for most stations and a numeric string for the rest.
- **Short Name:** public station identifier used for joins with Citi Bike trip data.
- **Name:** station name.
- **Latitude** and **Longitude:** station coordinates.
- **Capacity:** number of available docks.
- **From station_status:**
 - **Bikes available:** currently rentable bikes.
 - **Vehicle types available:** per-type counts of the bikes currently at the station.
 - **Docks available:** currently available docks.
 - **Operational flags:** `is_installed`, `is_renting`, and `is_returning`.

2.1.2 Extracted Features

Using the three operational flags (`is_installed`, `is_renting`, and `is_returning`), we derive a boolean `active` indicator to filter out unavailable stations from live views.

2.2 Analysis and Insights

The analysis focuses on station capacity, live availability, and docks reconciliation.

Capacity Distribution Figure 6 illustrates network capacity provisioning, which distinguishes smaller stations from major hubs. Most facilities cluster tightly around 20–30 docks, meanwhile the bar at 0 docks is not corrupt data: those are stations that appear in the feed but are not active (demonstrated in the [Jupyter notebook](#)).

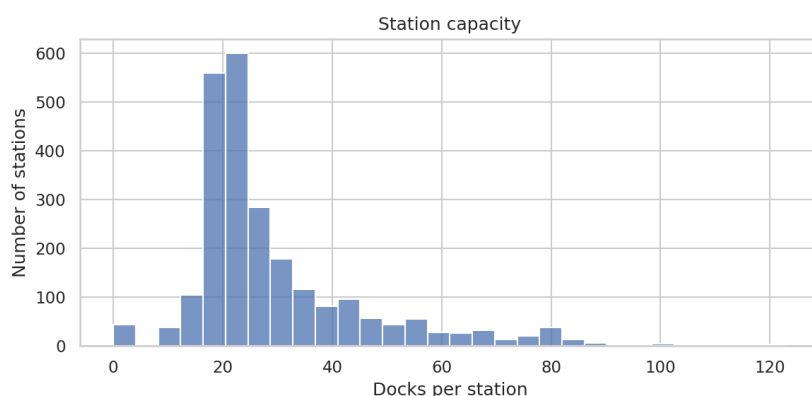


Figure 6: Distribution of station capacity, measured as the number of docks per station.

Live Availability Live bike and dock availability reflects the real-time state of the system. Because the feed is a dynamic snapshot, Figure 7 represents a single

point in time rather than a strictly reproducible state. The chart decomposes total network capacity, classifying every dock into exactly one of six defined states. The same was done for active and inactive stations, for a more granular view.

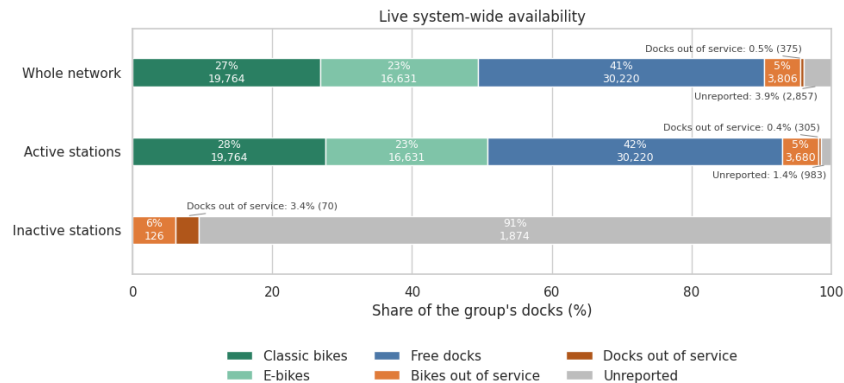


Figure 7: System-wide capacity breakdown across all, active, and inactive stations. Total capacity is fully decomposed into six mutually exclusive states: rentable classic bikes, rentable e-bikes, free docks, out-of-service bikes, out-of-service docks, and unreported capacity.

By capturing these dynamics, the breakdown demonstrates how the system can power user-facing tools with accurate, real-time availability insights.

2.3 Data Quality

The main quality issues concern unreported capacity, the self-consistency of the feed, and identifier consistency. Ultimately, as demonstrated in the notebook, there are no missing values for stations in the feed.

Inactive Stations The feed includes stations being installed, repaired, or removed, all grouped under the `inactive` category. They make up 101 out of 2,462 stations (4.1%) and 2,043 docks (2.8% of capacity).

Figure 7 shows why we remove them, since they report capacity but no live counts, and their bikes are out of service. Keeping them would show false empty stations on the map and distort capacity stats without adding real bike availability.

Unreported Capacity The missing values (Figure 7) arises from using two different endpoints. Station capacity is a static property provided by `station_information` and updated periodically, whereas live bike and dock counts are delivered continuously by `station_status`. Figure 8 highlights this discrepancy, showing the per-station gap between declared capacity and the sum of live counters.

Filtering out the inactive stations described above eliminates the main source of variance without altering real availability. For any remaining active stations that do not reconcile, dock states are read directly from live counters, treating capacity as a rough benchmark.

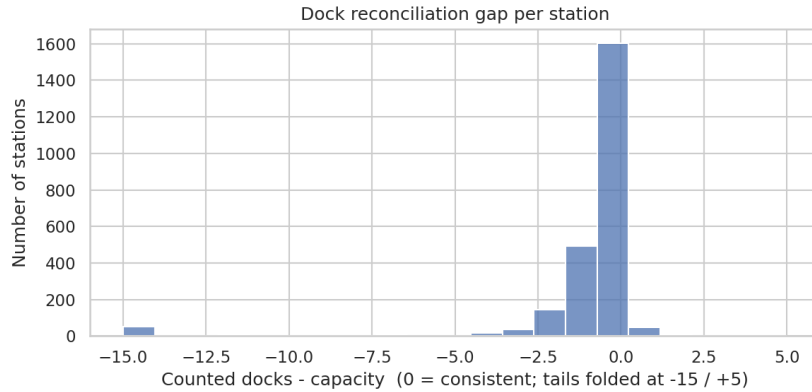


Figure 8: Per-station gap between the declared capacity and the docks accounted for by the live counters.

Identifier Consistency GBFS identifies stations in two ways: an internal `station_id` (either a UUID or a numeric string) and a public `short_name`. Trip history files use only the latter: 93.8% of station references in historical trips match a `short_name`, whereas none match a `station_id`. Therefore, all joins between live metadata and trip history should rely on the `short_name` field.

Live-Only Metadata Citi Bike publishes station metadata only as a live feed, with no archive of past station states. Linking trips to it therefore describes a past month through the network as it stands today. Stations removed since then keep their trips but lose every feed attribute, while stations added since appear with no history, which is easily mistaken for disuse. The bias is confined to what the feed contributes, capacity and operational state, since each trip already carries its own station name and coordinates.

3 Weather Dataset

Our second data source is hourly NYC weather, retrieved from the Open-Meteo archive API.¹

3.1 Dataset Structure

The dataset contains hourly weather observations for New York City, aligned to the `America/New_York` timezone. It provides environmental variables used to explain changes in cycling demand.

3.1.1 Original Structure

The dataset includes the following main fields:

- **Timestamp:** hourly observation time.
- **Temperature (`temperature_2m`):** air temperature in °C.
- **Precipitation:** hourly precipitation in millimetres.
- **Weather Code:** WMO weather condition code.

¹<https://open-meteo.com/>

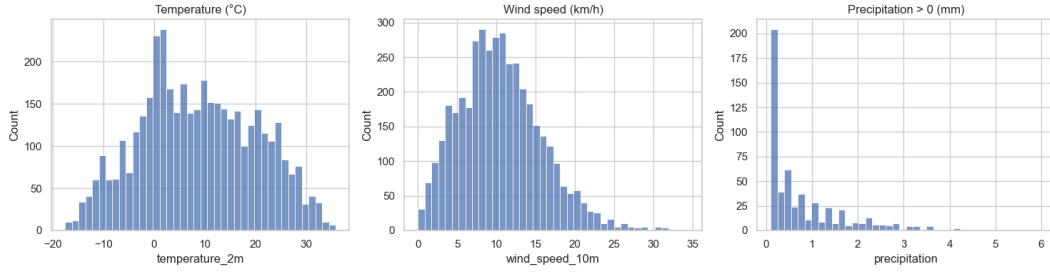


Figure 9: Temperature, wind speed and precipitation distributions over time.

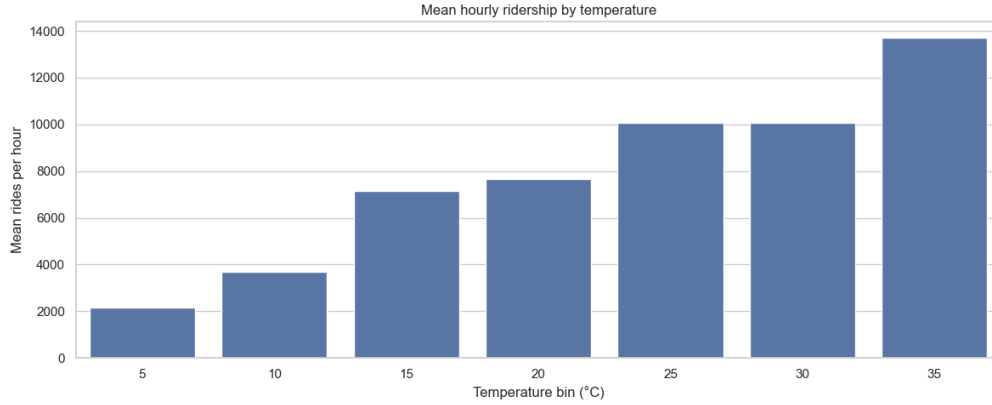


Figure 10: Ridership vs. temperature.

- **Wind Speed (`wind_speed_10m`):** wind speed in km/h.
- **Year:** derived field used to partition the dataset into annual Parquet files.

The dataset is used almost entirely in its original form, with only the **Year** column added during preprocessing. Weather data are retrieved from the Open-Meteo archive for the same period covered by the trip dataset.

3.2 Analysis and Insights

The exploratory analysis focuses on weather distributions and their relationship with ridership.

Temperature follows a clear annual cycle, while precipitation is highly right-skewed: most hours record no rainfall, with occasional heavy events. Consequently, precipitation histograms focus on non-zero observations. Wind speed shows moderate variation throughout the year. Figure 9 summarizes the main weather distributions and their seasonal behaviour.

Hourly trip counts are joined with weather observations using their timestamps. The resulting analysis shows that ridership, as expected, is higher during warmer periods and lower during cold or inclement weather. Two plateaus are observed between 15°C and 20°C, and between 25°C and 30°C. With higher temperatures, ridership increases noticeably, as shown in Figure 10.

These results justify including weather as an explanatory variable for temporal variations in bike usage.

3.3 Data Quality

The archived weather data are complete and consistently sampled at hourly intervals, with no significant missing values.

The main limitation is spatial coverage: the dataset represents weather at a single observation location rather than capturing microclimate differences across the five boroughs. While this may introduce some bias, it is a common approximation for city-scale analyses.

Precipitation is strongly skewed, so non-zero values are analysed separately from dry hours. The derived **Year** column is used only for data partitioning, while additional features such as temperature bins are created only when required for specific analyses.

4 Bike Lanes Dataset

Our third data source is NYC’s bike-route network, published by NYC OpenData² as dataset `mzxcg-pwib`. It provides the city’s cycling infrastructure and forms the infrastructure layer of the final visualization.

4.1 Dataset Structure

The dataset describes New York City’s bicycle infrastructure at the segment level. Each row represents a physical bike-lane segment, including its geometry, location, status, installation history, and facility type.

4.1.1 Original Structure

The original CSV contains one row per route segment and includes the following relevant attributes:

- **Segment ID**: unique identifier of the route segment.
- **Bike ID**: identifier used to track the segment across revisions.
- **Status**: whether the segment is active or retired.
- **Installation Date**: date the segment entered service.
- **Retirement Date**: date the segment was removed, if applicable.
- **the_geom**: line geometry used for map rendering.
- **Street, From Street, To Street**: street location.
- **Facility Class**: type of cycling infrastructure.
- **Borough**: borough containing the segment.

The analysis derives summaries by borough, status, and facility class. Unused fields (`prevbikeid`, `gwsys2`, `spur`, `ft2facilit`, and `tf2facilit`) were removed during preprocessing.

²<https://data.cityofnewyork.us/d/mzxcg-pwib>

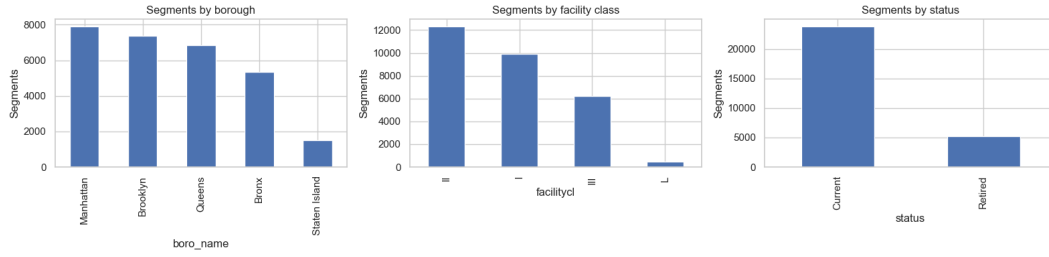


Figure 11: Breakdown of bike-lane segments by borough, facility class and status.

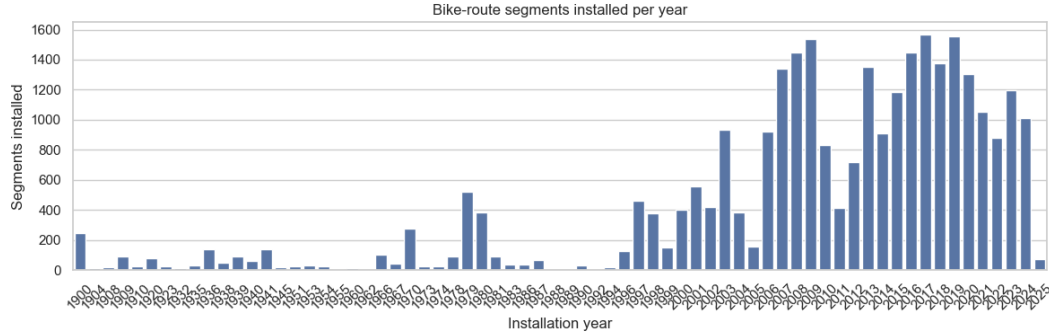


Figure 12: Bike-lane installation timeline.

4.2 Analysis and Insights

The analysis focuses on four aspects that directly support the final visualization.

A breakdown by borough shows that bike-lane segments are concentrated in Manhattan and Brooklyn, while the remaining boroughs contain fewer segments. A second breakdown compares segment status, showing that most infrastructure is currently active. Figure 11 summarizes the distribution of segments by borough, facility class, and status.

The facility-class distribution distinguishes protected lanes from shared facilities and is used to style the network in the visualization.

The installation timeline illustrates the gradual expansion of the bike network over time, reflecting continued investment in cycling infrastructure. Figure 12 shows the number of installed segments over the years.

4.3 Data Quality

The dataset is generally clean but requires limited preprocessing.

Unused columns were removed to simplify the analysis without affecting the visualization.

Current and retired segments are treated separately so that obsolete infrastructure is not included in the active network layer.

Installation and retirement dates are occasionally missing, causing slight underestimation in historical summaries. When the installation date is unknown, NYC OpenData records the placeholder value 01/01/1900 instead of leaving it blank, so this sentinel year appears as the lower bound of the historical timeline.

Records with missing or invalid geometries cannot be displayed and are filtered

before mapping. For the interactive visualization, only a sample of segments is rendered to improve responsiveness while preserving the overall spatial pattern.